7. Data Quality Audit

Audit summary

- Records: 100,000; fields: 37.
- Missing cells: 9,511 (0.257% of all cells).
- Complete duplicate rows: 0.
- Duplicate `vehicle_id` values: 0.
- Numeric fields contain wide ranges; extreme values should be reviewed with source definitions rather than automatically deleted.

Missingness by field

Field | Missing | Records (%)

`resale_value_pct_3yr` | 3,057 | 3.06%

`battery_cycle_life` | 2,028 | 2.03%

`customer_rating` | 1,931 | 1.93%

`infotainment_screen_inch` | 1,506 | 1.51%

`annual_sales_units` | 989 | 0.99%

Selected range checks

Field | Minimum | Maximum | Audit note

model_year | 2018 | 2026 | 2018–2026 observed; confirm valid coverage

battery_capacity_kwh | 1.50 | 410.00 | Maximum is unusually high; verify units/records

range_km | 55.00 | 659.90 | Review extremes against vehicle class

efficiency_wh_per_km | 9.98 | 2,242.84 | Maximum is much higher than median; inspect

price_usd | $900.04 | $409,995.49 | Wide distribution; confirm currency and vehicle classes

annual_sales_units | 10.00 | 772,442.00 | Strong high-end outlier; validate before mean-based comparison

customer_rating | 1.75 | 5.00 | Scale source not provided

safety_rating_stars | 1 | 5 | Observed 1–5; 15 records have one star

Assessment

Missingness is concentrated in `battery_cycle_life` (2,028), `resale_value_pct_3yr` (3,057), `annual_sales_units` (989), and `customer_rating` (1,931). No duplicate IDs were found, so `vehicle_id` is unique in the supplied file.

Recommended handling

1. Do not impute ratings or sales without a defensible rule.
2. For a metric-specific chart, exclude nulls only from that metric and state the effective record count.
3. Investigate extreme values using source records and units.

4. Keep an untouched raw file; save any cleaned version separately.
5. Recalculate all summary statistics after any transformation.